Bank Account Smart Contract on a Test Network

Operations: Deposit, Withdraw, Show balance | Solidity ^0.8.20 | Remix IDE, Remix VM (simulated test blockchain)

Step 1: Smart contract code

```
pragma solidity ^0.8.20;

contract BankAccount {

    mapping(address => uint256) private balances;

    event Deposited(address indexed customer, uint256 amount, uint256 newBalance);
    event Withdrawn(address indexed customer, uint256 amount, uint256 newBalance);

    function deposit() external payable {
        require(msg.value > 0, "Deposit must be greater than 0");

        balances[msg.sender] += msg.value;

        emit Deposited(msg.sender, msg.value, balances[msg.sender]);
    }

    function withdraw(uint256 amount) external {
        require(amount > 0, "Withdraw amount must be greater than 0");
        require(balances[msg.sender] >= amount, "Insufficient balance");

        balances[msg.sender] -= amount;

        (bool success, ) = payable(msg.sender).call{value: amount}("");
        require(success, "Transfer failed");

        emit Withdrawn(msg.sender, amount, balances[msg.sender]);
    }

    function showBalance() external view returns (uint256) {
        return balances[msg.sender];
    }

    function balanceOf(address customer) external view returns (uint256) {
        return balances[customer];
    }
}
```

BankAccount.sol (comments removed)

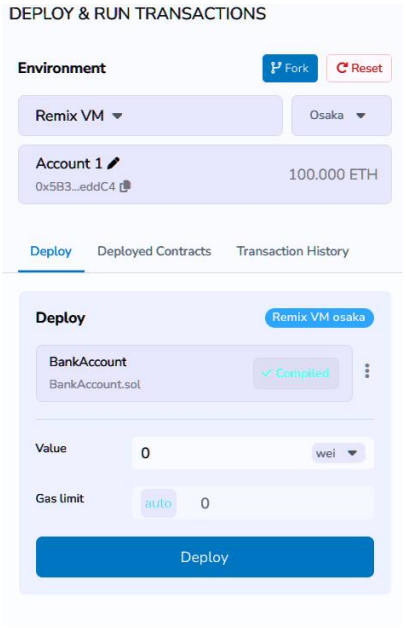


Fig 1: Deploy panel: BankAccount compiled, Remix VM, Account 1 has 100 test ETH

Step 2: Contract file in the Remix workspace

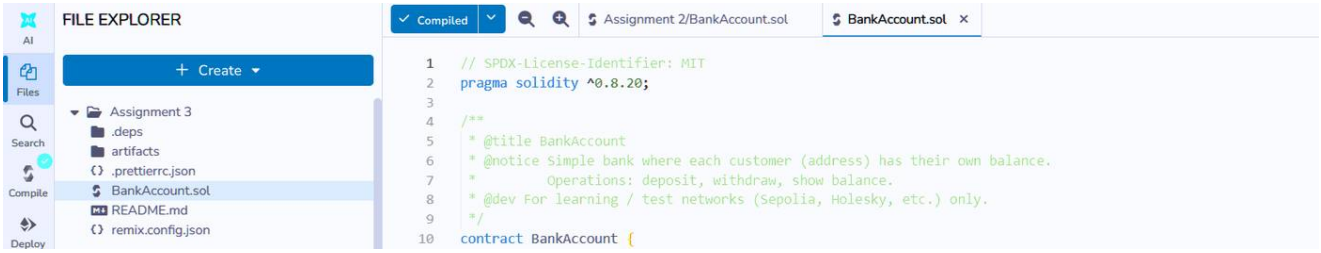


Fig 2: BankAccount.sol in the Assignment 3 workspace

Step 3: Compiled with the Solidity compiler

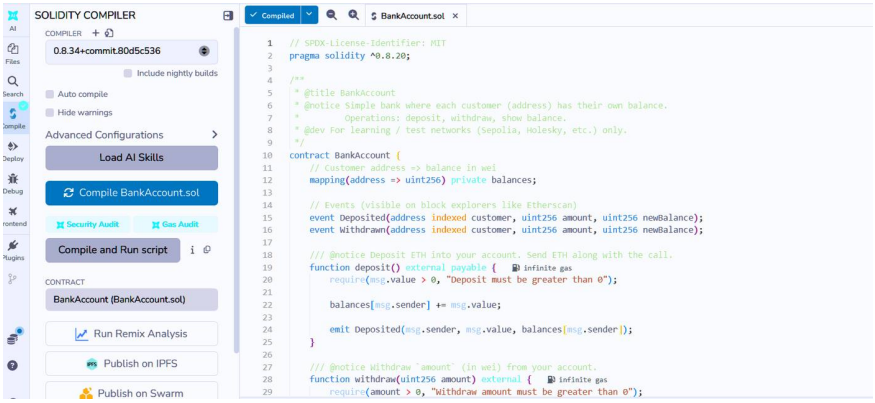


Fig 3: Solidity compiler 0.8.34, BankAccount.sol compiled successfully

Step 4: Contract deployed



Fig 4: BankAccount deployed on Remix VM at 0xd91...39138 (constructor transaction in the console)

Step 5: Deposit money

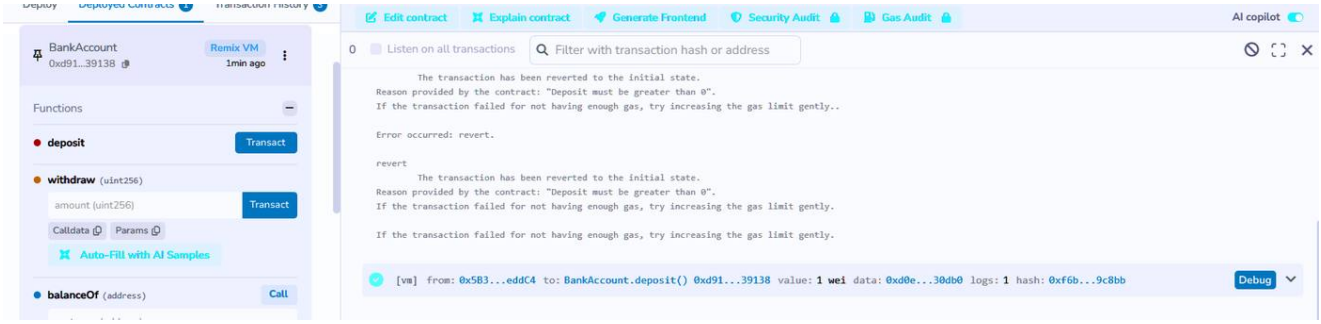


Fig 5: deposit() with 1 wei succeeds (green). Depositing 0 is rejected: "Deposit must be greater than 0"

Step 6: Show balance



Fig 6: Balance call returns 0:uint256: 1 (1 wei) for Account 1

Step 7: Withdraw money

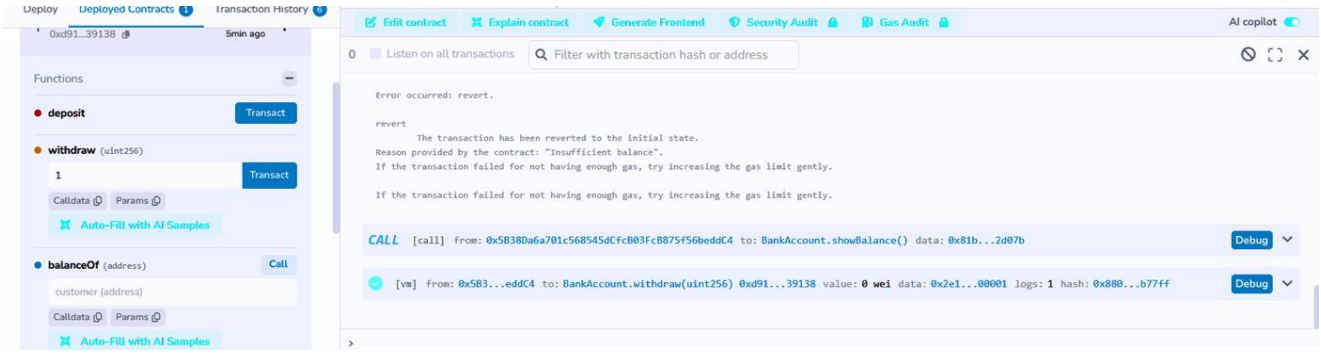


Fig 7: Withdrawing more than the balance is rejected: "Insufficient balance". After checking the balance, withdraw(1) succeeds

Step 8: All operations in sequence

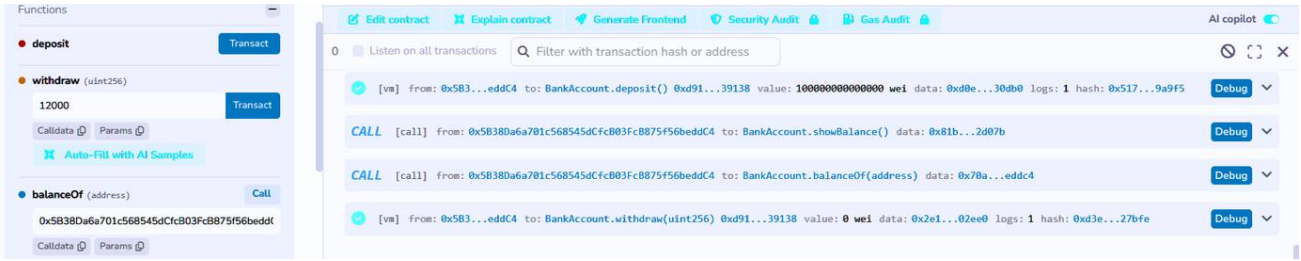


Fig 8: deposit of 100000000000000 wei, showBalance, balanceOf, then withdraw of 12000 wei, all successful

Summary

Operation	Function	Result observed
Deposit money	deposit() payable	Success; 0 wei deposit reverted with message
Withdraw money	withdraw(uint256 amount)	Success; amount above balance reverted
Show balance	showBalance(), balanceOf(address)	Returned the stored balance in wei

Conclusion: The BankAccount contract keeps a separate balance for each customer address. Deposit adds the sent ETH, withdraw sends it back only if the balance is sufficient, and the balance can be viewed at any time.